
Radial Velocity and Size of Scrape-Off-Layer Filaments Across 294 MAST Discharges from 100 kHz Visible Imaging

Jessie Dong
Stanford University

Abstract

Filaments carry much of the cross-field transport in the tokamak scrape-off layer, and models predict how their radial velocity scales with their size. We measure both on MAST from the 100 kHz, 48×256 pixel strip recordings of the midplane fast camera in the public FAIR-MAST archive. An automatic pipeline built from standard computer-vision operations (running-minimum background, robust thresholding, connected components, Hungarian linking, least-squares tracks, and optical flow) produces 86,042 filament tracks in 294 discharges across four campaigns. The archive distributes no spatial calibration for these recordings; we recover the pixel scale for each camera configuration by regressing the image column of the plasma limb on the EFIT outboard LCFS radius across discharges, with bootstrap intervals, and obtain 2.7–5.2 mm per pixel. The median radial velocity is 424 m/s and the median radial width is 3.6 cm. After correction for exposure blur, the velocity–width relation has a log–log slope of +0.05 (95% interval +0.03 to +0.07), so the velocity is nearly independent of size over the measured range. In two-region units the filaments span $\hat{a} = 0.13$ –1.97, straddling the model’s velocity maximum, which accounts for the flat dependence; but their normalized velocity is $\hat{v} \approx 0.06$, far below the model scale. Code and derived data are released.

1 Introduction

Filaments are field-aligned structures of excess density and temperature that form near the last closed flux surface (LCFS) of a tokamak and move radially outward through the scrape-off layer (SOL). They carry a large part of the cross-field particle and heat flux in the SOL (Krasheninnikov, 2001; Zweben et al., 2007; D’Ippolito et al., 2011), and their radial velocity sets how far they travel before they drain to the divertor targets, which in turn sets the load on the first wall. Analytic models give the velocity as a function of filament size and of the parameters that control how the polarization charge of the filament is dissipated. When the charge drains through the sheath at the target, the velocity falls with size, $v \propto a^{-2}$ (Krasheninnikov, 2001); when ion polarization current closes the circuit inside the filament, the velocity rises with size, $v \propto a^{1/2}$ (Garcia et al., 2006). The two-region model of Myra et al. (2006) joins these limits and predicts a velocity maximum at a critical size a^* set by the sound speed, the connection length, and the collisionality.

Testing these predictions needs velocities and sizes for many filaments under known plasma conditions. On the Mega Amp Spherical Tokamak (MAST) the main instrument for this has been a Photron SA1.1 fast camera that views the outboard midplane in visible light. Earlier MAST studies used wide-angle views, identified filaments by hand or with detection software built for that view, and worked from a small number of discharges (Dudson et al., 2008; Ben Ayed et al., 2009; Kirk et al., 2016; Farley et al., 2019). Each needed a camera calibration model fitted to the view in use.

The FAIR-MAST archive (Jackson et al., 2024) makes the full MAST data set public. It includes a recording mode of the midplane camera that earlier work did not use at scale: a 48×256 pixel strip through the outboard edge recorded at 100,000 frames per second. The strip resolves radial motion at 10 μ s intervals but spans only 48 rows, so it constrains radial

motion and radial size and little else. The archive does not include a spatial calibration for these recordings, so the pixel scale has to be recovered from the data.

This paper turns those recordings into filament statistics. We build an automatic pipeline from standard computer-vision operations: a running-minimum background model, robust thresholding, connected-component detection, Hungarian frame-to-frame assignment, least-squares track fitting, and Lucas–Kanade optical flow as an independent velocity estimate. We calibrate the pixel scale against the equilibrium reconstruction: the image column of the plasma limb moves with the EFIT outboard LCFS radius from shot to shot, and a robust regression over a campaign gives the scale in millimetres per pixel with a bootstrap confidence interval. We run the pipeline on every usable strip recording in the archive.

The contributions are:

- Radial velocity and radial width for 86,042 filament tracks in 294 MAST discharges, selected automatically from the public archive. The median radial velocity is 424 m/s and the median radial full width at half maximum is 3.6 cm; 81% of tracks move outward.
- A pixel-scale calibration per camera configuration that uses only the images and EFIT, with a stated uncertainty, and an account of what that uncertainty does and does not allow.
- A test of the blob velocity scalings. The pooled velocity–size slope is +0.09 in log–log units and +0.05 after the widths are corrected for exposure blur, so the velocity is nearly independent of size over the measured range. In two-region units the filaments sit at $\hat{a} \approx 0.13$ – 1.97 , where the model places its velocity maximum, but their normalized velocity $\hat{v} \approx 0.06$ is far below the model scale.
- Dependences of per-shot median velocity and width on plasma current, Greenwald fraction, and beam power, with confidence intervals from robust regression.

Code and derived data accompany the paper.

2 Background

Filament motion. A filament is a field-aligned tube of plasma, with a cross-field radius a between a few ion gyroradii and a few centimetres, that is denser and hotter than the surrounding SOL. In the outboard SOL the magnetic curvature and gradient drifts separate ions and electrons across the filament. The resulting electric field drives an $E \times B$ drift that moves the whole filament radially outward (Krasheninnikov, 2001; D’Ippolito et al., 2011). The drift speed depends on the potential the charge separation can build, and so on the paths through which the polarization charge is dissipated. Two limits are standard. If the filament is connected along the field line to the divertor target, the charge drains through the sheath and $v \propto c_s (L_{\parallel}/R) (\rho_s/a)^2$, where c_s is the sound speed, ρ_s the ion sound gyroradius, $L_{\parallel}$ the connection length, and R the major radius (Krasheninnikov, 2001). If the ion polarization current closes the circuit inside the filament, $v \propto c_s (a/R)^{1/2}$ (Garcia et al., 2006).

The two-region model. Myra et al. (2006) model the midplane and the divertor leg as two coupled regions and join these limits. With

$$a^* = \rho_s^{4/5} L_{\parallel}^{2/5} R^{-1/5}, \quad v^* = c_s \left(\frac{2a^*}{R} \right)^{1/2}, \quad \Theta = \left(\frac{a}{a^*} \right)^{5/2}, \quad \Lambda = \frac{v_{ei} L_{\parallel}}{\Omega_e \rho_s}, \quad (1)$$

and $\hat{a} = a/a^*$, $\hat{v} = v/v^*$, the model gives

$$\hat{v} = \begin{cases} \hat{a}^{1/2} & \text{connected ideal interchange } (\Lambda < 1, \Theta < 1), \\ \hat{a}^{-2} & \text{sheath connected } (\Lambda < 1, \Theta > 1), \\ \Lambda \hat{a}^{-2} & \text{resistive X-point } (1 < \Lambda < \Theta), \\ \hat{a}^{1/2} & \text{resistive ballooning } (\Lambda > 1, \Lambda > \Theta). \end{cases} \quad (2)$$

Here ν_{ei} is the electron-ion collision frequency and Ω_e the electron gyrofrequency (D’Ippolito et al., 2011). At low collisionality the velocity rises as $\hat{a}^{1/2}$ for small filaments, peaks near $\hat{a} = 1$, and falls as $\hat{a}^{-2}$ for large ones. The model is a scaling argument: the prefactors depend on the filament profile and on the parallel structure, and simulations in realistic geometry find velocities below the model values by factors of order unity (Easy et al., 2014; Walkden et al., 2015; Paruta et al., 2018). Theiler et al. (2009) measured the velocity-size relation on the TORPEX basic plasma device and recovered the predicted transitions between regimes.

What a camera measures. Visible-light imaging records line-integrated D_α emission, which depends on the local density, temperature, and neutral density. Filaments appear as bright, field-aligned streaks against the quiescent background; their emission tracks the density perturbation closely enough that velocities and widths from imaging agree with probe measurements within tens of percent (Zweben et al., 2007; Ben Ayed et al., 2009). In a tangential view the brightest image feature is the plasma limb, the band where the line of sight is tangent to the flux surfaces near the LCFS. Filaments cross the limb as they move outward. Because the camera integrates over each exposure, a filament that moves a distance $v t_{\text{exp}}$ during the exposure is smeared by that distance along its direction of motion. We treat this effect explicitly in Section 6.

3 Data

Archive. FAIR-MAST (Jackson et al., 2024) serves the MAST data set as one Zarr store per discharge over HTTP, with a metadata API for discovering signals and shot parameters. The archive covers 15,969 discharges. We read six groups from it: the midplane fast camera (rbb), the EFIT equilibrium reconstruction (efm), the summary signals for line-averaged density and Greenwald density (esm), the interferometer (ane), the core and edge Thomson scattering systems (ayc, aye), and the midplane D_α monitor (xim). Plasma current, heating scheme, injected beam power, and the flat-top interval come from the per-shot metadata table.

Camera. The rbb signal is the Photron SA1.1 fast camera (“bullet camera B”) at the HM10 midplane port, which views the outboard edge tangentially (Kirk et al., 2016; Farley et al., 2019). In the recording mode we use, the camera reads out a 48×256 pixel region of its sensor at 100,000 frames per second with 10-bit depth; the region offset on the sensor is recorded with each shot. Exposure times are 1–10 μs , with 4, 7, and 10 μs the common settings. The metadata view string records the lens, aperture, and filter in use and whether the camera was mounted inverted; the SOL-side detection in Section 4 makes the pipeline independent of this orientation. Appendix A lists the camera configurations.

Selection. 468 discharges have a 48×256 rbb recording. Each recording is a burst at the fast frame rate embedded in a slow record, at 1–2 kHz, of the whole discharge; the burst lasts 100 ms in the median (range 3–500 ms), and in 419 recordings its frame interval is 10 μs . We read the time base of every recording and keep a discharge if its burst overlaps the plasma flat top, extended by 20 ms on each side, by at least 2,000 fast frames (375 discharges), its flat top is defined and at least 50 ms long (404), and its peak plasma current is at least 400 kA (421). 368 discharges satisfy all three. For each we download the burst frames inside the extended flat top; 368 discharges were downloaded and processed, with a median window of 186 ms and 3.5 million frames in total. 16 of these have a burst frame rate of 50 kHz rather than 100 kHz; we process them but keep them out of the statistics so that the time resolution is uniform. Table 1 summarizes the processed set by campaign.

Equilibrium and plasma parameters. From EFIT, reconstructed every 5 ms, we take at each time the LCFS boundary contour, q_{95} , the plasma current, the magnetic axis position, the vacuum toroidal field at the axis, and the minor radius. The outboard LCFS radius is the largest major radius of the boundary contour within 5 cm of the axis height, which is the quantity the camera limb tracks. Line-averaged density comes from the summary signal

Campaign	Years	Shots	In stats	Exp. (μs)	Pooled scale (mm/px)	Own scales	$\bar{v}_r$ (m/s)	$\bar{\delta}_r$ (cm)
M6	2006–2007	257	220	1, 5, 7, 10	3.7 (3.3–4.1)	3 (2.9–3.7)	419	3.6
M7	2008–2009	15	13	4, 10	3.3 (3.0–3.8)	1 (3.3)	451	3.6
M8	2011	32	25	2, 4	5.2 (3.6–11.5)	1 (5.2)	627	5.2
M9	2013	48	36	4, 10	2.9 (2.5–4.2)	1 (2.7)	323	2.8

Table 1: Processed discharges by campaign. “In statistics” counts discharges with at least 50 selected tracks. The scale is the calibration of Section 4.6 with its 95% bootstrap interval. $\bar{v}_r$ and $\bar{\delta}_r$ are the medians over discharges of the per-discharge median radial velocity and width.

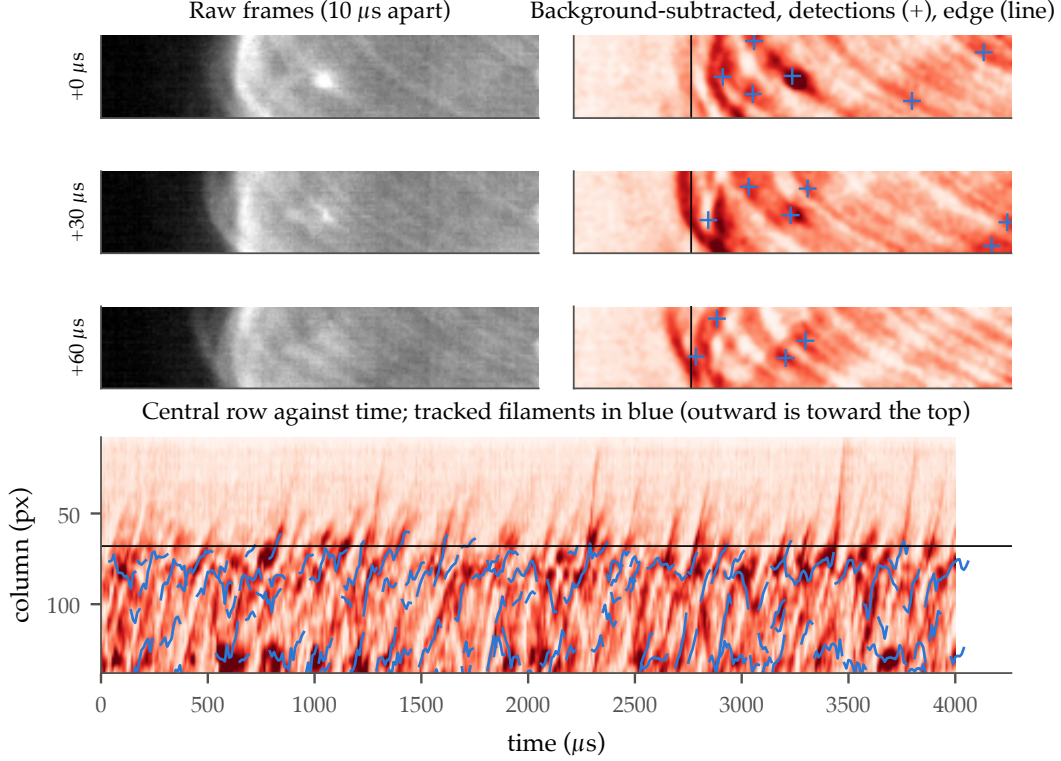


Figure 1: Pipeline stages on discharge 15235 (M6, 7 μs exposure). Top: three raw frames 30 μs apart and the corresponding background-subtracted frames with detections (crosses) and the edge column (line). Bottom: the central row of the background-subtracted strip against time, with tracked filaments overlaid. Filaments appear as inclined streaks that move toward the SOL (upward in this view).

where it exists and from the CO₂ interferometer summary otherwise; Greenwald fraction uses $n_{\text{GW}} = I_p / (\pi a^2)$ with I_p in MA and a the EFIT minor radius. Thomson profiles exist for the later campaigns only; for 60 of the 294 discharges in the statistics a profile reaches the LCFS during the flat top.

4 Method

Figure 1 shows the stages on one discharge. Every stage runs with the same settings on every shot; there is no per-shot tuning.

4.1 Strip geometry and the plasma edge

A recording is an array of n frames of 48×256 pixels with 10-bit depth. The columns run across the plasma edge: the SOL occupies one end of the strip and the core the other, and which end depends on how the camera was mounted in that campaign. We take the row-averaged brightness profile $p(t, x)$, smooth it over 200 frames (2 ms) in time and $\sigma = 2$ columns in space, and call the darker outer quarter of the columns the SOL side. This fixes a sign $s = \pm 1$ such that outward motion is $s dx/dt$. The edge column $x_{\text{half}}(t)$ is the first column, scanning from the SOL side, at which p exceeds the midpoint between the dark level (the mean of the ten outermost SOL-side columns) and the interior level (the median over the columns 40–75% of the way across the strip from the SOL side), with linear sub-pixel interpolation. The limb peak $x_{\text{peak}}(t)$ is the maximum of p within 40 columns inboard of x_{half} , refined by parabolic interpolation. Detections are referenced to x_{half} during processing and shifted to x_{peak} at aggregation, because the peak is the feature that the calibration in Section 4.6 ties to the equilibrium.

4.2 Background subtraction and detection

Filaments are transient, so a per-pixel running minimum over 21 frames (± 10 frames, $\pm 100 \mu\text{s}$) estimates the quiescent background (Kirk et al., 2016). The fluctuation image is the frame minus this background, passed through a 3×3 median filter and a Gaussian with $\sigma = 1$ pixel. The noise level σ_n is 1.4826 times the median absolute deviation of all fluctuation pixels in a block of 3,000 frames. Pixels above $3\sigma_n$ are labelled into 8-connected components, and components smaller than 12 pixels are dropped. For each remaining component we record the intensity-weighted centroid (x_c, y_c) , the peak fluctuation, the orientation and the major and minor axis lengths from the second central moments of the component, and the radial width. The radial width is measured on a chord: we average the fluctuation over the five rows centred on y_c and take the number of columns at or above half the chord maximum. We use a chord rather than the full component because filaments appear as inclined streaks, and the column extent of the whole component grows with the tilt. The signed SOL distance of a detection is $s(x_c - x_{\text{half}})$.

4.3 Tracking

Detections in consecutive frames are linked by the Hungarian method (Kuhn, 1955) with the Euclidean distance between centroids as the cost and a gate of 12 pixels per frame, which is about 4.5 cm or 4.5 km/s at the calibrated scale. Unmatched detections start new tracks, and a track ends at the first frame without a match. Tracks with fewer than five detections (50 μs) are discarded. For each track the radial velocity in pixels per frame is the least-squares slope of $s x_c$ against frame index, with a standard error from the residuals; the width is the median chord FWHM over the track and the SOL distance is the median signed distance. Recordings are processed in blocks of 3,000 frames with a 60-frame overlap. Tracks that start in the overlap are dropped from the earlier block and recovered in the later one.

4.4 Optical flow as an independent velocity

As a check on the tracking, we also estimate the velocity at each detection by Lucas–Kanade optical flow (Lucas & Kanade, 1981): Sobel spatial gradients and the frame difference in a 7×7 window around the centroid give a 2×2 least-squares system for the displacement between frames f and $f + 1$, which we solve when its condition number is below 10^4 . The track-level flow velocity is the mean of the horizontal component over the track’s detections, with the sign s applied. The flow estimate uses no assignment step and is sensitive to the brightness profile rather than the centroid, so agreement between the two is evidence that neither is an artefact of its method.

4.5 ELM detection

Edge-localized modes (ELMs) eject large filaments that would dominate a velocity distribution. We detect them from the midplane D_α monitor, sampled at 50–250 kHz depending on the campaign: the residual after a 5 ms running median is a spike where it exceeds six times its robust standard deviation, and each spike’s time is the centre of mass of its samples. The monitor is coarsely digitized in some campaigns, with only tens of distinct values, which makes the robust standard deviation vanish; the threshold is therefore floored at six digitizer steps and at half the inter-ELM level. Because the D_α channel is absent or saturated in some discharges, we apply the same detector to the frame-mean camera brightness. A track that begins within 1 ms of a spike of either kind is flagged, and discharges with more than five D_α spikes inside the camera window are labelled ELMy. Flagged tracks are excluded from all statistics; the parameter dependences in Section 5 use the non-ELMy discharges only.

4.6 Pixel scale from the equilibrium

The archive gives no spatial calibration for the strip recordings, so we recover the scale from the data. The limb peak sits where the line of sight is tangent to the flux surfaces at the LCFS, so its sensor column should move with the outboard LCFS radius R_{LCFS} from EFIT, and the slope of column on radius is the inverse pixel scale. Within a single discharge R_{LCFS} moves by only a few centimetres during the flat top, the EFIT reconstruction is least reliable during ramp-up and termination, and the brightness profile changes shape as the density rises, so a within-shot regression is unstable (Appendix B). Between discharges, the plasma position varies by 10–20 cm while the camera and its optics stay fixed through a campaign, which gives a much longer lever arm.

For each discharge we take, at every EFIT time inside the flat top, R_{LCFS} as the largest major radius of the EFIT boundary contour within 5 cm of the magnetic-axis height, and x_{peak} from the profile smoothed over 2 ms. EFIT samples whose R_{LCFS} differs by more than 8 cm from the flat-top median are dropped as unconverged. The discharge then contributes one point: its median x_{peak} , converted to sensor coordinates by adding the region-of-interest offset recorded in the camera metadata, and its median R_{LCFS} . We call a camera configuration the combination of campaign, view string in the camera metadata, and region-of-interest offset, because a change in any of these can mean a change of lens, pointing, or sensor region. A configuration with at least 8 discharges and at least 3 cm of spread in R_{LCFS} gets its own Theil–Sen slope (Sen, 1968), which we keep if its sign agrees with s , the Spearman correlation satisfies $|\rho| \geq 0.5$, and the bootstrap interval excludes zero. Every other configuration uses the pooled slope of its campaign, fitted on all the campaign’s configurations after subtracting each configuration’s mean R_{LCFS} and mean column, which removes the per-configuration intercepts. The 95% intervals come from 2,000 bootstrap resamples of discharges within configurations (Efron & Tibshirani, 1993). The scale is the inverse of the slope magnitude.

4.7 Physical units and track selection

A track’s radial velocity is $v_r = s \dot{x}_c \times f_{\text{frame}} \times \ell$ with ℓ the pixel scale, and its radial width is $\delta_r = \text{FWHM} \times \ell$. We measure the SOL distance from x_{peak} , using the per-shot median offset $x_{\text{peak}} - x_{\text{half}}$. A track enters the statistics if its median position lies between 10 pixels inboard and 40 pixels outboard of the limb peak (3–5 cm inboard to 11–21 cm outboard, depending on the configuration’s scale), its velocity standard error is below 0.5 pixels per frame, its width is finite, it begins inside the plasma flat top, and it is not flagged as ELM-adjacent. A discharge enters the shot-level statistics if it has at least 50 such tracks with $v_r > 0$.

4.8 Statistics

All central values are medians with 95% bootstrap intervals from 2,000 resamples. Slopes are Theil–Sen estimates in $\log_{10}$ – $\log_{10}$ coordinates with the estimator’s own 95% interval, and we report Spearman’s ρ alongside. The velocity–size slope is computed per discharge and also pooled: for the pooled fit we divide each track’s v_r and δ_r by the medians of its discharge, so that differences between discharges in the pixel scale or the plasma conditions do not create a spurious correlation. To correct for exposure blur we replace δ_r by $\delta_{\text{corr}} = (\delta_r^2 - (v_r t_{\text{exp}})^2)^{1/2}$, which treats the smear as a Gaussian of width $v_r t_{\text{exp}}$ added in quadrature, and refit. Parameter dependences are Theil–Sen slopes of the per-discharge median v_r and δ_r on plasma current, Greenwald fraction, and injected beam power, the last offset by 0.05 MW so that Ohmic discharges enter the log fit.

For the two-region normalization we need T_e and n_e at the LCFS, B , q_{95} , and R . T_e comes from the Thomson scattering profile, from the edge system where it exists and the core system otherwise, interpolated at R_{LCFS} and taken as the median over the flat top; when no profile reaches the LCFS we assume $T_e = 30$ eV and say so. We take $n_{e,\text{sep}}$ as half the line-averaged density, B as the vacuum toroidal field at the magnetic axis scaled by $R_{\text{axis}}/R_{\text{LCFS}}$, and $L_{\parallel} = \pi q_{95} R_{\text{axis}}$. The filament radius in the model is half the measured FWHM, $a = \delta_r/2$. v_{ei} uses the NRL formulary (Huba, 2019) with deuterium ions.

5 Results

5.1 Pixel scale

Figure 2 shows the between-shot calibration and Table 2 in the appendix lists every camera configuration. Of the 23 configurations, 6 have enough discharges and enough spread in R_{LCFS} for their own fit, and they cover 241 of the 352 processed discharges; the rest use the pooled slope of their campaign. The limb-peak column falls linearly with R_{LCFS} in every fitted configuration, with Spearman correlations between -0.58 and -0.94 , and the sign of every slope agrees with the SOL direction found from the images. The fitted scales span 2.7–5.2 mm per pixel. The tightest fit is the M6 configuration “HM10, image normal, f/1.8” (70 discharges, $\rho = -0.94$), at 2.9 mm per pixel (2.7–3.0). The M9 configuration gives 2.7 mm per pixel (2.5–3.4). For the same camera, port, and campaign, Farley et al. (2019) quote a resolution of 5 mm at the tangency plane for wide-angle images reduced to 256×160 pixels from the camera’s 640×448 mode, which is about 2 mm per native pixel. Our M9 value is of that order, although 2 mm lies below its interval, and the archive does not record whether the lens was the same.

Two features of the calibration limit what the physical units can support. First, the intervals are 10–30% wide, and a bias common to a configuration, such as a drift of the emission peak relative to the LCFS with density, is not covered by them. Second, in pixel units the filaments look alike in every configuration: across the 6 fitted configurations with at least ten discharges the median track velocity is 1.15–1.40 pixels per frame and the median width is 8–13 pixels, while the fitted scales differ by a factor of 1.9. The differences between campaigns in Table 1 are therefore the differences between their calibrations, and we do not interpret them. A scale error multiplies v_r and δ_r of a configuration by the same factor; it cancels in the velocity–size slopes of Section 6, which are fitted after normalizing each discharge by its own medians, and it moves a discharge along a diagonal in the regime diagram without changing the conclusions drawn there.

5.2 Velocity and width distributions

294 discharges have at least 50 selected tracks and enter the statistics, with 86,042 tracks in total. Figure 3 shows the pooled distributions. The median radial velocity over discharges is 424 m/s (95% interval 406–439) and the median radial width is 3.6 cm (3.4–3.7). The velocity distribution is skewed, with a tail to 2–3 km/s; the width distribution peaks at 2–3 cm and has a tail to 10 cm. Among all near-edge tracks that pass the quality cuts, 81% move outward; the inward-moving remainder is concentrated inboard of the limb peak,

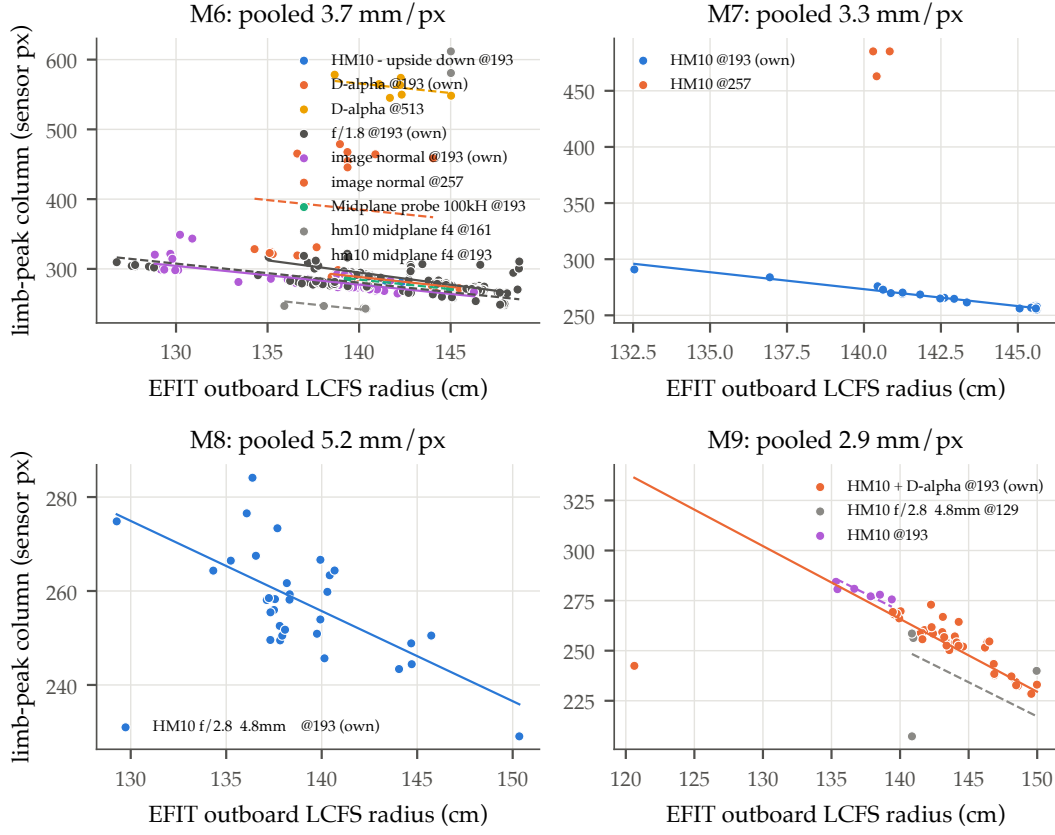


Figure 2: Pixel-scale calibration. Each point is one discharge: the median limb-peak column in sensor coordinates over the flat top against the median EFIT outboard LCFS radius. Colours are camera configurations (view string and region-of-interest offset); solid lines are a configuration's own Theil-Sen fit and dashed lines the pooled campaign slope with the configuration's intercept. The scale is the inverse of the slope.

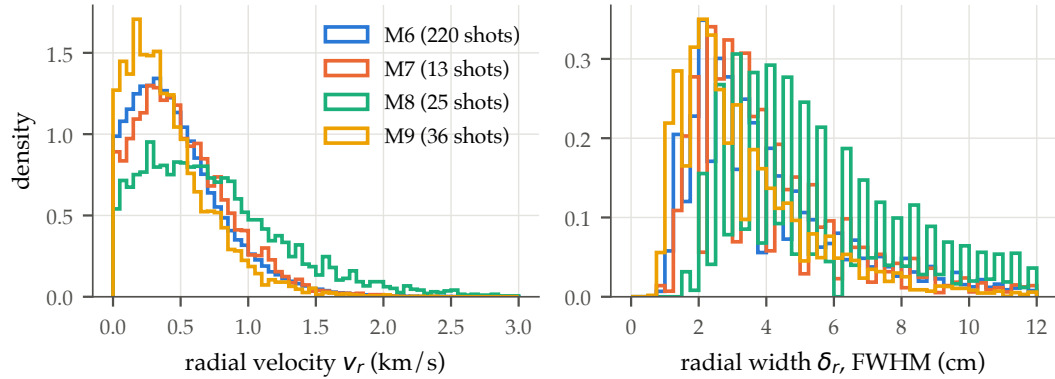


Figure 3: Distributions of radial velocity (left) and radial FWHM (right) of selected tracks, by campaign.

where the field-aligned streaks sweep across the strip as they rotate with the plasma. We keep only outward-moving tracks in the velocity-size analysis.

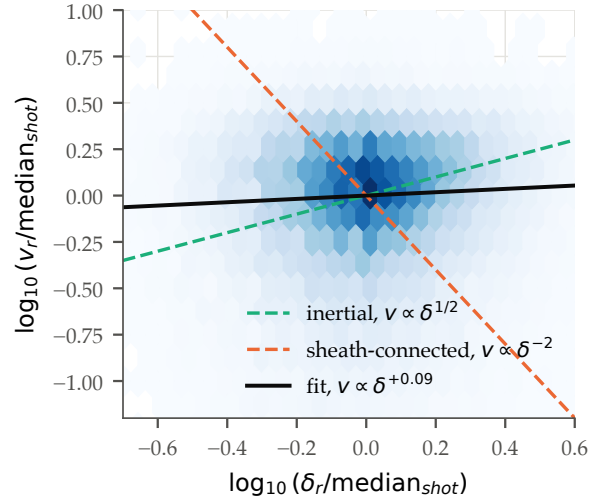


Figure 4: Radial velocity against radial width, each normalized by the median of its discharge, for all selected tracks (hexagonal bin density). The solid line is the pooled Theil–Sen fit; dashed lines are the sheath-connected and inertial scalings through the medians.

5.3 Independent velocity estimates

Two estimates that do not use the tracking confirm the velocity scale. The cross-correlation lag between the row-averaged fluctuation profiles of consecutive frames inside the near-edge window, with no detection step, gives a per-discharge median velocity of 257 m/s over the 175 discharges in which more than half the frame pairs have a usable correlation peak. Its ratio to the tracking median is 0.63 (0.52–0.68), and the two correlate across discharges with Spearman $\rho = +0.29$. The lag method weights every bright structure in the window, including the slow inboard part of the streaks and the stationary limb, so a ratio below one is expected. Lucas–Kanade flow at the detections correlates with the track velocities within discharges (median per-discharge correlation +0.23; $\rho = +0.49$ between discharge medians) but is smaller by a factor of 0.13. The streaks are elongated and inclined, so the brightness-constancy constraint fixes only the component of motion normal to the streak, and displacements exceed one pixel per frame; both bias the flow magnitude down. We therefore use the flow only as a sign and ordering check.

6 Analysis

6.1 Velocity against size

Figure 4 shows the velocity–size relation after each discharge is normalized by its own medians. The per-discharge Theil–Sen slope of $\log v_r$ on $\log \delta_r$ has a median of +0.11 (+0.08–+0.15) and is positive in 70% of discharges; the pooled slope is +0.09 (+0.06–+0.11) with Spearman $\rho = +0.05$. The relation is far from both limiting scalings: the sheath-connected δ^{-2} is excluded, and the inertial $\delta^{1/2}$ lies well above the interval.

6.2 Exposure blur

A filament that moves during the exposure is smeared along its direction of motion, and faster filaments are smeared more, which produces a positive velocity–width correlation with no physical content. The median smear $v_r t_{\text{exp}}$ is 0.31 cm, about a tenth of the median width, so the effect is small but not negligible. Two tests bound it. First, the slope depends on exposure in the expected direction within M6, where the two common settings share the optics: the per-discharge slope has a median of +0.05 (+0.01–+0.11) at 7 μs ($n = 105$) and +0.16 (+0.11–+0.27) at 10 μs ($n = 142$), while the median width is the same, 3.4 against

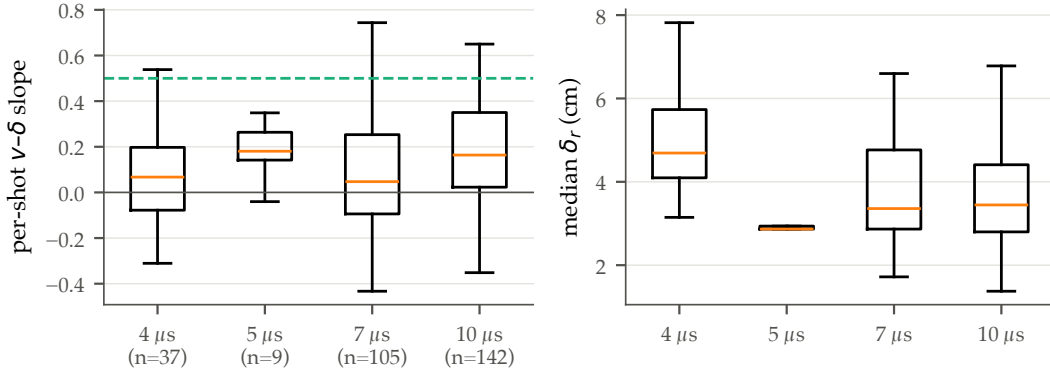


Figure 5: Effect of exposure time. Left: per-discharge velocity–width slopes by exposure setting; the dashed line is the inertial scaling. Right: per-discharge median width by exposure.

3.4 cm, because the extra smear of 3 μ s at 400 m/s is 0.1 cm. The 4 μ s discharges belong to M7 and M8 and are not comparable in width, but their slope, +0.07 (-0.04+0.16), is also low. Second, correcting each width for its own smear in quadrature lowers the pooled slope to +0.05 (+0.03+0.07). We take this corrected value as the measurement: the radial velocity is nearly independent of width over the measured range, with at most a weak positive dependence.

6.3 Two-region normalization

With the inputs of Section 4, the median reference scales are $a^* = 2.6$ cm, $v^* = 7345$ m/s, and $\Lambda = 0.3$, so the discharges sit in the low-collisionality half of the regime diagram. Figure 6 places each discharge at its median $\hat{a} = (\delta_r/2)/a^*$ and $\hat{v} = v_r/v^*$. The filaments span $\hat{a} = 0.13$ –1.97 (median 0.69), which straddles the velocity maximum of the model; over the central half of this range the model’s $\hat{v}(\hat{a})$ stays within a factor of two of its peak. A sample that straddles the peak is expected to show a weak net dependence of velocity on size, and this is what we measure. The normalized velocity, however, is $\hat{v} \approx 0.06$: the measured filaments move at a few percent of the model’s velocity scale. Part of this gap is the input assumptions, since $v^* \propto T_e^{7/10}$ and T_e at the LCFS is assumed for 234 of the 294 discharges, and part is the model itself, which gives the velocity of an isolated, fully formed filament with an order-unity prefactor that simulations in realistic geometry find to be below one (Walkden et al., 2015; Paruta et al., 2018). The measurement is also made within 15 cm of the limb, where filaments are still forming and accelerating. The ratio is nonetheless larger than the prefactor uncertainty alone explains, and we regard the amplitude, not the size dependence, as the point of tension with the model.

6.4 Dependence on plasma parameters

Figure 7 shows the per-discharge medians against plasma current and Greenwald fraction for the 183 non-ELMy discharges, which span $I_p = 402$ –1127 kA and $f_{GW} = 0.14$ –1.01. No dependence is resolved. The log–log slope of the median velocity on plasma current is -0.05 (-0.29+0.19) and on Greenwald fraction -0.06 (-0.18+0.05); for the width the slopes are +0.10 (-0.21+0.40) and +0.06 (-0.08+0.21). Beam power has no effect on the velocity (-0.01, -0.03+0.01) and a small positive one on the width (+0.05, +0.01+0.07). Because these fits pool configurations with different calibrations, we repeat them inside the best-calibrated configuration alone (“HM10, image normal, f/1.8”, 37 non-ELMy discharges at one scale, $I_p = 606$ –1127 kA): the velocity slopes are -0.29 (-1.02+0.66) on current and -0.10 (-0.26+0.12) on Greenwald fraction, consistent with the pooled result. Kirk et al. (2016) found in a dedicated current scan at fixed density and power that the radial velocity falls with current. Our sample is not a controlled scan: current, density, heating, and campaign vary

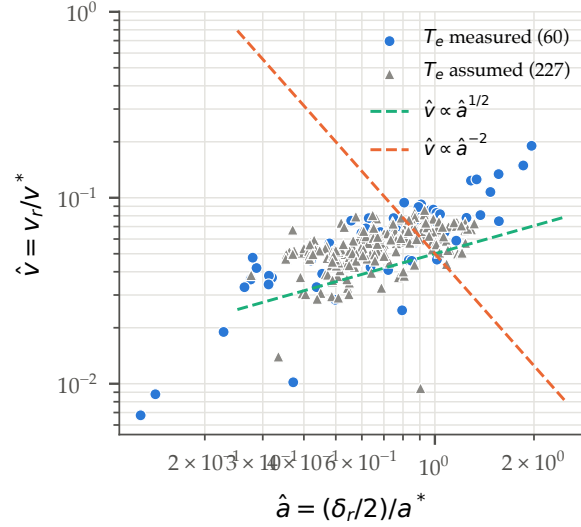


Figure 6: Per-discharge median velocity and size in two-region units. Dashed lines are the inertial ($\hat{v} \propto \hat{a}^{1/2}$) and sheath-connected ($\hat{v} \propto \hat{a}^{-2}$) branches. Circles use a measured T_e at the LCFS; triangles assume $T_e = 30$ eV.

together, and most discharges sit near 700 kA, so a weak dependence is within the reach of confounding. We report the slopes as they are and do not claim a contradiction.

7 Related Work

Filament imaging on MAST. [Dudson et al. \(2008\)](#) used the midplane fast camera with automated analysis of L-mode images and BOUT simulations. They found that the toroidal mode number of the edge turbulence rises with density and falls with q_{95} , that filament width has the opposite dependence, and that simulated filaments are born within about 2 cm of the edge in a self-generated $E \times B$ shear layer. [Ben Ayed et al. \(2009\)](#) combined camera images with a reciprocating Langmuir probe during inter-ELM periods and showed that filaments rotate near the LCFS and move outward, with more and faster filaments at higher density; they used the radial velocities to test the sheath-connected and inertial scalings. [Kirk et al. \(2016\)](#) scanned the plasma current at fixed density and heating power in L-mode and found that the radial velocity, and to a lesser extent the radial size, decrease as the current rises, and that this dependence is consistent with the measured divertor profiles. [Farley et al. \(2019\)](#) introduced a tomographic inversion of wide-angle images onto a basis of equilibrium field lines, which turns filament identification into a problem in field-aligned coordinates. All of these studies relied on a camera geometry model built with Calcam ([Silburn et al., 2018](#)) for the view in use. The present work differs in three ways: it uses the 48-row strip mode at 100 kHz rather than wide-angle frames, it recovers the pixel scale from the equilibrium instead of a geometry model, and it processes hundreds of discharges across four campaigns rather than a scan of a few.

Blob theory and simulation. The sheath-connected scaling is due to [Krasheninnikov \(2001\)](#) and the inertial scaling to [Garcia et al. \(2006\)](#). [Myra et al. \(2006\)](#) introduced the two-region model that we use for normalization; [D’Ippolito et al. \(2011\)](#) review the model and its comparisons with experiment. [Theiler et al. \(2009\)](#) measured the velocity–size relation on TORPEX and recovered the predicted regime transitions. Three-dimensional simulations of isolated filaments ([Easy et al., 2014](#); [Walkden et al., 2015](#); [Paruta et al., 2018](#)) find that the scalings hold in form but that realistic geometry and parallel dynamics change the prefactors. [Zweben et al. \(2007\)](#) review edge turbulence imaging across devices, including the relation between light emission and density fluctuations that imaging measurements rely on.

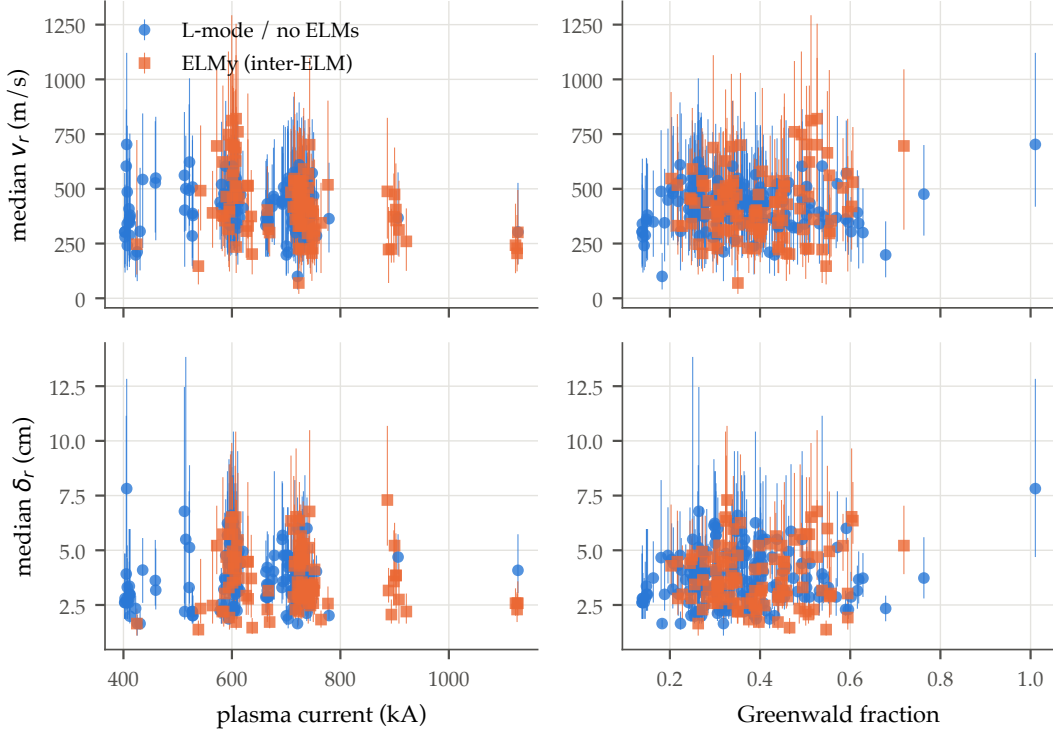


Figure 7: Per-discharge median radial velocity (top) and width (bottom) against plasma current (left) and Greenwald fraction (right). Bars are interquartile ranges over tracks. Circles are discharges without ELMs in the camera window; squares are ELMy discharges, measured between ELMs.

Methods. The detection and tracking steps are standard: connected-component labelling after robust thresholding, frame-to-frame assignment by the Hungarian method (Kuhn, 1955), and Lucas–Kanade optical flow (Lucas & Kanade, 1981). Slopes use the Theil–Sen estimator (Sen, 1968) and confidence intervals use the bootstrap (Efron & Tibshirani, 1993). The data come from the FAIR-MAST archive (Jackson et al., 2024).

8 Limitations

Calibration. The pixel scale rests on the assumption that the limb peak tracks the EFIT outboard LCFS radius with a fixed offset within a camera configuration. The offset can drift with the emission profile, which depends on density and neutral pressure, and EFIT’s boundary carries its own uncertainty of order a centimetre. The bootstrap intervals in Table 2 include the scatter these effects produce but not a common bias. The near-uniformity of the filaments in pixel units across configurations whose fitted scales differ by almost a factor of two (Section 5) is either a coincidence or a sign that part of the spread in the fitted scales is not optical; we cannot decide this from the archive, and we have therefore drawn no conclusion that depends on comparing physical units between configurations. Every velocity and width in the paper should be read with a systematic uncertainty of 10–30% in its configuration’s scale, and a comparison between configurations with up to a factor of two. A per-shot Calcam model would remove this uncertainty and is the natural next step if the calibration images can be released. Appendix B shows why the within-shot regression is not a substitute.

Geometry. The strip gives a single row band through the edge. The column axis is close to radial only at the tangency plane, and the width we measure is the chord length of an

inclined streak, which overestimates the width perpendicular to the filament by $1/\cos\theta$ with θ the streak angle to the column axis. We do not correct for this, because the angle is poorly determined in a 48-row strip; the effect inflates δ_r by a factor that is roughly constant within a campaign and therefore shifts $\hat{a}$ but not the velocity–size slope. Emission is line-integrated along the view, so a bright structure behind the tangency plane projects to a smaller apparent radius; this blurs the SOL distance but not the velocity, which is a time derivative.

What the camera sees. D_α emission depends on neutral density as well as plasma density, and the neutral density falls steeply inside the LCFS. Filaments are therefore visible mainly in the SOL, and a filament’s brightness changes as it moves. Tracking by centroid is robust to a slowly changing amplitude; optical flow is not, which is one reason the flow velocities are low.

Selection. We measure between 10 pixels inboard and 40 pixels outboard of the limb, that is from 3–5 cm inboard to 11–21 cm outboard depending on the configuration, a region that includes filaments at birth. Velocities in the far SOL may differ. The 50-track threshold per discharge and the exclusion of ELM-adjacent tracks remove some discharges and some of the largest events; the headline numbers describe inter-ELM and L-mode filaments near the edge.

Model inputs. T_e at the LCFS is assumed for most discharges, and $n_{e,\text{sep}}$ is taken as half the line average. Both enter a^* , v^* , and Λ as power laws with exponents below one, so a factor of two in either moves a discharge by less than a factor of two in the regime diagram. The conclusion that the sample straddles $\hat{a} = 1$ is robust to this; the value of $\hat{v}$ is not better than a factor of two.

9 Conclusion

Public archives make it possible to measure filament statistics on hundreds of discharges with a pipeline that no one tunes by hand. On MAST, the 100 kHz strip recordings give radial velocities and widths for 86,042 filaments in 294 discharges, with a pixel scale recovered from the equilibrium for each camera configuration to 10–30%. The radial velocity has a median of 424 m/s and does not depend on filament width to within a log–log slope of about 0.1 once exposure blur is removed. In two-region units the filaments sit around the predicted velocity maximum, which explains the flat dependence, but they move at a few percent of the model’s velocity scale. The amplitude discrepancy, the campaign-level calibration, and the radial-only geometry are the three places where a dedicated experiment or a released geometry model would improve on what the archive alone can give.

Acknowledgements

I thank the professors who teach computer vision at Stanford, notably Juan Carlos Niebles, Adrien Gaidon, and Silvio Savarese; their courses led me to the ideas behind this work.

References

- N. Ben Ayed, A. Kirk, B. Dudson, S. Tallents, R. G. L. Vann, H. R. Wilson, and the MAST Team. Inter-ELM filaments and turbulent transport in the Mega-Amp Spherical Tokamak. *Plasma Physics and Controlled Fusion*, 51(3):035016, 2009. doi: 10.1088/0741-3335/51/3/035016.
- D. A. D’Ippolito, J. R. Myra, and S. J. Zweben. Convective transport by intermittent blob-filaments: Comparison of theory and experiment. *Physics of Plasmas*, 18(6):060501, 2011. doi: 10.1063/1.3594609.

-
- B. D. Dudson, N. Ben Ayed, A. Kirk, H. R. Wilson, G. Counsell, X. Xu, M. Umansky, P. B. Snyder, B. Lloyd, and the MAST Team. Experiments and simulation of edge turbulence and filaments in MAST. *Plasma Physics and Controlled Fusion*, 50(12):124012, 2008. doi: 10.1088/0741-3335/50/12/124012.
- L. Easy, F. Militello, J. Omotani, B. Dudson, E. Havlíčková, P. Tamain, V. Naulin, and A. H. Nielsen. Three dimensional simulations of plasma filaments in the scrape off layer: A comparison with models of reduced dimensionality. *Physics of Plasmas*, 21(12):122515, 2014. doi: 10.1063/1.4904207.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.
- T. Farley, N. R. Walkden, F. Militello, M. Sanna, J. Young, S. S. Silburn, J. Harrison, L. Kogan, I. Lupelli, S. S. Henderson, A. Kirk, and J. W. Bradley. Filament identification in wide-angle high speed imaging of the Mega Amp Spherical Tokamak. *Review of Scientific Instruments*, 90(9):093502, 2019. doi: 10.1063/1.5109470.
- O. E. Garcia, N. H. Bian, and W. Fundamenski. Radial interchange motions of plasma filaments. *Physics of Plasmas*, 13(8):082309, 2006. doi: 10.1063/1.2336422.
- J. D. Huba. NRL plasma formulary. Naval Research Laboratory, Washington, DC, 2019.
- Samuel Jackson, Saiful Khan, Nathan Cummings, James Hodson, Shaun de Witt, Stanislas Pamela, Rob Akers, Jeyan Thiyagalingam, and the MAST Team. FAIR-MAST: A fusion device data management system. *SoftwareX*, 27:101869, 2024. doi: 10.1016/j.softx.2024.101869.
- A. Kirk, A. J. Thornton, J. R. Harrison, F. Militello, N. R. Walkden, the MAST Team, and the EUROfusion MST1 Team. L-mode filament characteristics on MAST as a function of plasma current measured using visible imaging. *Plasma Physics and Controlled Fusion*, 58(8):085008, 2016. doi: 10.1088/0741-3335/58/8/085008.
- S. I. Krasheninnikov. On scrape off layer plasma transport. *Physics Letters A*, 283(5–6): 368–370, 2001. doi: 10.1016/S0375-9601(01)00252-3.
- H. W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
- Bruce D. Lucas and Takeo Kanade. An iterative image registration technique with an application to stereo vision. In *Proceedings of the 7th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 674–679, 1981.
- J. R. Myra, D. A. Russell, and D. A. D’Ippolito. Collisionality and magnetic geometry effects on tokamak edge turbulent transport. I. a two-region model with application to blobs. *Physics of Plasmas*, 13(11):112502, 2006. doi: 10.1063/1.2364858.
- Paola Paruta, C. Beadle, P. Ricci, and C. Theiler. Blob velocity scaling in diverted tokamaks: a comparison between theory and simulation. Technical Report WP14ER-PR(18) 21313, EUROfusion, 2018. Preprint.
- Pranab Kumar Sen. Estimates of the regression coefficient based on Kendall’s tau. *Journal of the American Statistical Association*, 63(324):1379–1389, 1968.
- S. Silburn, J. Harrison, M. Smithies, A. Wynn, T. Farley, and J. Cavalier. Calcam (version 1.9.4). <https://github.com/euratom-software/calcam>, 2018.
- C. Theiler, I. Furno, P. Ricci, A. Fasoli, B. Labit, S. H. Müller, and G. Plyushchev. Cross-field motion of plasma blobs in an open magnetic field line configuration. *Physical Review Letters*, 103(6):065001, 2009. doi: 10.1103/PhysRevLett.103.065001.
- N. R. Walkden, B. D. Dudson, L. Easy, G. Fishpool, and J. T. Omotani. Numerical investigation of isolated filament motion in a realistic tokamak geometry. *Nuclear Fusion*, 55(11): 113022, 2015. doi: 10.1088/0029-5515/55/11/113022.
-

S. J. Zweben, J. A. Boedo, O. Grulke, C. Hidalgo, B. LaBombard, R. J. Maqueda, P. Scarin, and J. L. Terry. Edge turbulence measurements in toroidal fusion devices. *Plasma Physics and Controlled Fusion*, 49(7):S1–S23, 2007. doi: 10.1088/0741-3335/49/7/S01.

A Camera configurations

Table 2 lists the camera configurations in the processed set: the view string from the camera metadata, the sensor column offset of the region of interest, the number of discharges, the spread of R_{LCFS} across the discharges, the Spearman correlation between limb-peak column and R_{LCFS} , the configuration’s own scale where it could be fitted, and whether the configuration uses its own scale or the pooled campaign scale (Section 4.6).

Camp.	View string	ROI left	Shots	ΔR_{LCFS} (cm)	ρ	Scale (mm/px)	Used
M6	HM10, normal, f/1.8	193	70	13	-0.94	2.9 (2.7–3.0)	own
M6	HM10, normal	193	61	17	-0.89	3.7 (3.1–4.1)	own
M6	hm10 midplane f4	193	59	22	-0.47	3.9 (3.6–5.1)	pooled
M6	HM10, normal, D α 10 nm	193	30	7	-0.88	3.5 (3.2–3.9)	own
M6	HM10, normal	257	14	10	+0.34	0.9 (0.4–1.9)	pooled
M6	HM10, normal, D α 10 nm	513	7	6	–	–	pooled
M6	hm10 midplane f4	161	6	4	–	–	pooled
M6	HM10 - upside down	193	3	1	–	–	pooled
M6	probe 100kHz	193	3	6	–	–	pooled
M6	HM10, normal, D α 10 nm	545	2	0	–	–	pooled
M6	HM10, normal, D α 10 nm	225	1	0	–	–	pooled
M6	HM10, normal	225	1	0	–	–	pooled
M6	probe view	193	1	0	–	–	pooled
M7	HM10	193	22	13	-0.88	3.3 (3.0–3.8)	own
M7	HM10	257	3	1	–	–	pooled
M8	HM10 f/2.8 4.8mm nd 0.9	193	32	21	-0.58	5.2 (3.6–11.5)	own
M9	HM10 + D α filter	193	38	29	-0.82	2.7 (2.5–3.4)	own
M9	HM10	193	6	4	–	–	pooled
M9	HM10 f/2.8 4.8mm	129	4	9	–	–	pooled
M9	HM10 + D α filter	129	1	0	–	–	pooled
M9	HM10 + D α filter	513	1	0	–	–	pooled
M9	HM10 + D α filter	577	1	0	–	–	pooled
M9	HM10	129	1	0	–	–	pooled

Table 2: Camera configurations and their calibration. A configuration uses its own scale when it has at least 8 discharges, at least 3 cm of spread in R_{LCFS} , a correlation of the right sign with $|\rho| \geq 0.5$, and a bootstrap interval that excludes zero.

B Within-shot calibration

A within-shot regression of the limb-peak column on R_{LCFS} uses the few centimetres of plasma movement during a flat top. For the 326 discharges in which the LCFS moves by at least 2 cm, the implied scale has a median of 5.8 mm per pixel with an interquartile range of 3.8–13.6 and extremes below 1 and above 10^4 . The spread comes from three effects: the EFIT boundary is least reliable at the start and end of the flat top, where most of the movement occurs; the brightness profile changes shape as the density rises, which moves the peak at fixed R_{LCFS} ; and the lever arm is short, so a 1 cm error in R_{LCFS} changes the slope by tens of percent. The between-shot regression of Section 4.6 replaces a few centimetres of movement in one discharge with 10–20 cm of position variation across a campaign and is correspondingly more stable.

C ELM detection

Figure 8 shows the D_α trace and the camera frame-mean brightness for an ELMy discharge, with the detected spikes. The camera proxy finds the same events as the D_α monitor where both are available and serves as the only detector where the D_α channel is missing.

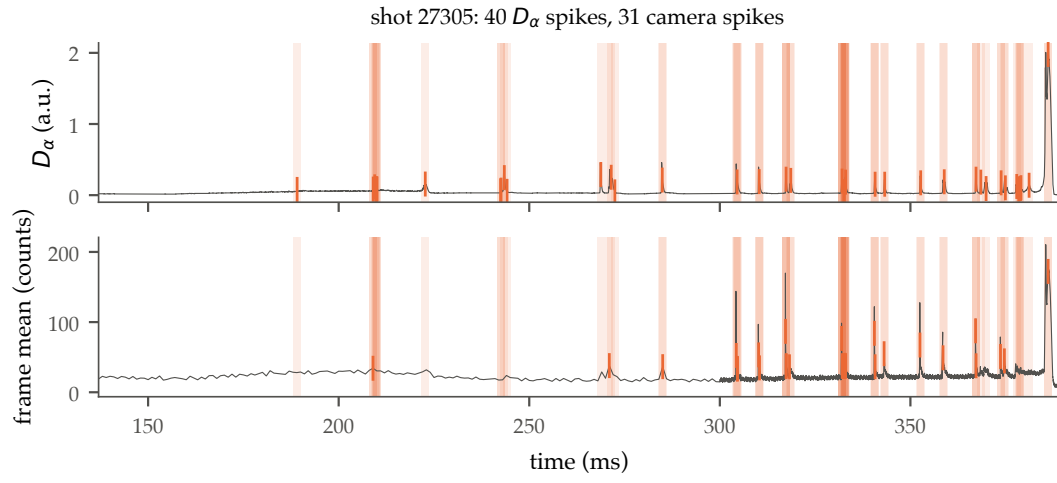


Figure 8: ELM detection on an ELMy discharge. Top: midplane D_α with detected spikes marked. Bottom: camera frame-mean brightness with its own spike detection. The shaded bands are the ± 1 ms exclusion windows.